EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2021
General Certificate of Education Advanced Level
Higher 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

23 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, civics group and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this question paper.
The use of an approved scientific calculator is expected, where appropriate.

1 Which electronic configuration represent an atom of an element, at ground state, that forms a stable ion with a charge of +3?

A $1s^2\ 2s^2\ 2p^3$

B $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

C $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

D $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^1$

Answer: B

A Has 5 valence e^-, thus it will gains 3 e^- to form a stable ion of -3 charge
B Has 3 valence e^-, thus it will lose 3 e^- to form a stable ion of +3 charge
C This is a group 18 element, and thus will not be able to form a stable ion of +3 charge.
D This electronic configuration is not at ground state. It should be $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$

2 Which particle has more protons than electrons and more protons than neutrons? (D = ${}^{2}_{1}H$)

A H_3O^+ **B** D_3O^+ **C** D_2O **D** OH^-

Answer: A

	H_3O^+	D_3O^+	D_2O	OH^-
Neutrons	8	3(1) + 8 = 11	2(1) + 8 = 10	8
Protons.	3(1) + 8 = 11	3(1) + 8 = 11	2(1) + 8 = 10	1 + 8 = 9
Electrons	11 – 1 = 10 Or 3+8-1	11 – 1 = 10 Or 3+8-1	2(1) + 8 = 10	9 + 1 = 10 Or 8+1+1

3 Gaseous particle **R** has a proton number n and forms a stable monoatomic ion of charge –1.

Gaseous particle **S** has a proton number of (n+2) and it forms a stable monoatomic ion which is isoelectronic with the ion of **R**.

Which of the following statement is correct?

A Ion of **S** has a smaller ionic radius than ion of **R**.

B Ion of **R** releases more energy than ion of **S** when an electron is added to each particle.

C **R** has a larger atomic radius than **S**.

D **S** requires less energy than **R** when an electron is removed from each particle.

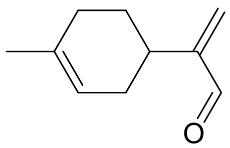
Answer: A

Since no. of protons in **R** = n and stable ion of **R** is **R^-**,
No. of electrons in **R** = n + 1.
In addition, since R forms a stable monoatomic ion of -1 charge, R is a group 17 element. E.g. C*l*.

As **S** is isoelectronic with **R**, no. of electrons in **S** = n + 1
Given than **S** has (n + 2) protons, thus ion of **S** is **S^+**.
Since R is from group 17, S will be from group 1 of the next period. E.g. K

A True. Both **S^+** and **R^+** are isoelectronic (**negligible difference** in shielding effect) but **S^+** has more protons and **R^+**, thus **S^+** will have a greater nuclear charge, and in turn greater effective nuclear charge, resulting in a smaller ionic radius.
B False. **R^-** takes in energy to form a doubly charged ion **R^{2-}** while **S^+** releases energy to form **S**.
C False. Despite the increase in nuclear charge, **S** has one more filled principal quantum shell than **R**, thus **S** has a larger atomic radius.
D False. **S^+** requires more energy to form **S^{2+}** than **R^-** to form **R**.

4 Limonene aldehyde is a fragrance additive in cosmetic products.



Which statements are true about the molecule?

1 Hydrogen bonds are formed between the molecules.

2 The molecule dissolves in organic solvent easily via instantaneous dipole–induced dipole interactions.

3 The bond angles about all carbon atoms in the molecule are the same.

A 1 and 2 **B** 2 and 3 **C** 2 only **D** 3 only

Answer: C

1: False. The aldehyde functional group is unable to form intermolecular hydrogen bond with another aldehyde group.
2: True. The non polar alkyl groups in limonene aldehyde forms instantaneous dipole–induced dipole interactions.
3. False. The carbon atoms of C=C and C=O is trigonal planar, at 120° while the rest of the carbon atoms are tetrahedral, at 109.4°.

5 In a single covalent bond, one pair of electrons is shared by two atoms. Which statement is true for **all** single covalent bond?

A Covalent bonds are formed when bonding electrons are shared equally between two atoms.

B Covalent bonds allow the atoms involved to complete their outer electron shell configuration.

C Covalent bonds formed between atoms of different electronegativities are stronger.

D Covalent bonds formed between overlap of larger and more diffused orbitals are weaker.

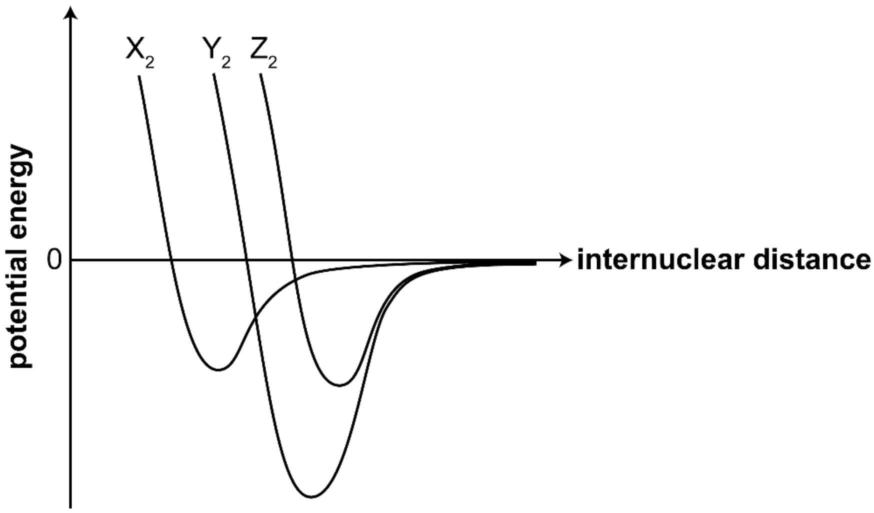
Answer: D

A: False. In polar bonds, bonding electrons are not centrally placed between the two atoms.
B: False. It is possible for atoms in a molecule to be electron deficient (fewer than 8 valence electrons) or to expand octet (more than 8 valence electrons)
C: False. There are non-polar bonds that are stronger than polar bonds. E.g BE(C-C) is 350 $kJmol^{-1}$ while BE(C-N) is 305 $kJmol^{-1}$.
D: True. Larger and more diffused orbitals overlap are less effective. Hence the bond formed is weaker.

6 *Use of the Data Booklet is relevant to this question.*

A Morse curve shows how the energy of a two atom system changes as a function of internuclear distance. The covalent bond length is determined by the distance at which the lowest potential energy is achieved. The lower the potential energy, the stronger the bond.

The Morse curve for three diatomic molecules, X_2, Y_2 and Z_2 are shown below.



Which of the following correctly identifies the diatomic molecules, X_2, Y_2 and Z_2?

X_2 Y_2 Z_2

A	H_2	N_2	O_2
B	H_2	O_2	N_2
C	N_2	O_2	H_2
D	O_2	H_2	N_2

Answer: A

H_2 has the smallest bond energy and longest bond length as the H-H single bond is weakest and the H atom is smaller.
N_2 has largest bond energy and hence shorter bond length than that of O_2 as there is triple bond formed between N atoms.

7 The melting point of lithium is 190 °C while that of magnesium is 639 °C. Which statement is more relevant in explaining this difference?

A The magnesium atom is larger than the lithium atom.

B The magnesium atom is less electronegative than the lithium atom.

C The magnesium ion has higher charge than the lithium ion.

D The magnesium ion contains more electrons than the lithium ion.

Answer: **C**

Metallic bond strength depends on both charge and ionic radius of the metal ion. Based on the options given the most relevant statement is that magnesium ion has a +2 charge while lithium ion has +1 charge.

8 Both carbon and silicon forms tetrachloride, CCl_4 and $SiCl_4$ respectively.

The tetrachlorides were added to water separately. Only the solution from $SiCl_4$ produced an acidic solution.

Which statements are true about the reaction of the above tetrachlorides with water?

1 SiO_2 formed from hydrolysis made the solution acidic.

2 $SiCl_4$ undergoes hydrolysis due to the vacant and energetically accessible d orbitals in Si.

3 CCl_4 is inert to hydrolysis as C−Cl bond is stronger that of Si−Cl.

A 1, 2 and 3 **B** 1 and 3 **C** 2 and 3 **D** 2 only

Answer: **D**

1: False. The acidic solution is due to formation of HCl. Although SiO_2 is an acidic oxide, it is not soluble in water.
2: True.

3. False. From Data Booklet, the bond energy of C−C*l* (340 $kJmol^{-1}$) and Si−C*l* (359 $kJmol^{-1}$) are similar. The inert nature of CCl_4 is due to the lack of vacant and energetically accessible d orbitals in C.

9 Two test tubes, each containing a sample of hydrogen chloride and hydrogen bromide, were heated. A colour change was observed for only one of the test tubes.

What best explain for the observation?

A HBr is more volatile than HC*l*.

B $Br_2(g)$ is more volatile than $Cl_2(g)$.

C HBr is a stronger reducing agent than HC*l*.

D H−Br bond is weaker than H−C*l* bond.

Answer: **D**

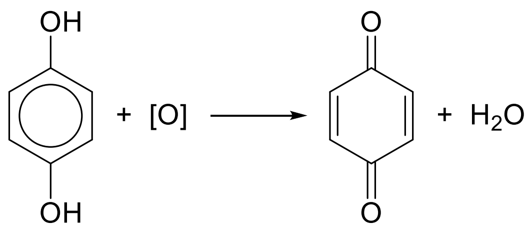
Reddish brown fumes of $Br_2(g)$ is observed.

This reaction describes the thermal decomposition of hydrogen halides.

$$HX(g) \rightarrow \tfrac{1}{2} X_2(g) + \tfrac{1}{2} H_2(g)$$

Down the group, the thermal stability of HX decreases due to weaker H-X bond as the orbitals of X gets larger, the overlap between H and X orbital becomes less effective.

10 A 1.0 g sample of **X** (M_r = 110.0) is oxidised to form **Y**.



X **Y**

[O] represents the oxidising agent used in the reaction.

What is the mass of **Y** formed from the above reaction?

A 0.964 g **B** 0.981 g **C** 1.00 g **D** 1.04 g

Answer: **B**

Amt of **X** $= \frac{1}{110} = 0.00909$ mol

= Amt of **Y**

Mass of **Y** = 0.00909 × (12 × 6 + 4 + 16 × 2) = 0.981

11 In the analysis of blood sample, an anticoagulant was added to form a complex ion, $[Ca(C_5H_6NO_4)_n]^{2-}$. The percentage by mass of the complex is: Ca, 12.2%; C, 36.6%; H, 3.7%; N, 8.5%; O, 39.0%.

What is the value of n?

A 1 **B** 2 **C** 3 **D** 4

Answer: **B**

	Ca	**C**	**H**	**N**	**O**
Mass in 100g / g	12.2	36.6	3.7	8.5	39.0
Ar	40.1	12	1	14	16
Amt / mol	0.3042	3.050	3.700	0.6071	2.438
ratio	1	10	12	2	8

The formula of the ion is $[Ca(C_{10}H_{12}N_2O_8)]^{2-}$.

Comparing with $Ca(C_5H_6NO_4)_n^{2-}$, n = 2

12 Sulfur dioxide gas decolourises acidified potassium manganate(VII) solution when bubbled through the solution, forming a sulfur-containing product.

In an experiment, 20.25 cm^3 of 0.0100 mol dm^{-3} potassium manganate(VII) was required to react with 12.15 cm^3 of sulfur dioxide gas at room temperature.

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

What is the identity of the sulfur-containing product?

A S^{2-} **B** SO_3^{2-} **C** $S_2O_6^{2-}$ **D** SO_4^{2-}

Answer: **D**

O.S. of S

in SO_2	+4
In S^{2-}	-2
In SO_3^{2-}	+4
In $S_2O_6^{2-}$	+5
In SO_4^{2-}	+6

Amt of MnO_4^- = $\frac{20.25}{1000} \times 0.01 = 0.0002025$ mol

Amt of SO_2 reacted = $\frac{12.15}{24000} = 0.0005063$ mol

Ratio of amt of MnO_4^- : amt of SO_2 = 2:5
Since $MnO_4^- \equiv 5e^-$ and $2MnO_4^- \equiv 5SO_2$, thus $2MnO_4^- \equiv 5SO_2 \equiv 10e^-$

1 mol of SO_2 loses 2 mol of e^- (oxidised) to form sulfur-containing compound where O.S. of S = +6 ⇒ SO_4^{2-}

13 Which of the following reactions represents the standard enthalpy change of formation of aluminium sulfate, $Al_2(SO_4)_3$?

A $2Al(s) + 3S(s) + 6O_2(g) \rightarrow Al_2(SO_4)_3(s)$

B $2Al(g) + 3S(g) + 12O(g) \rightarrow Al_2(SO_4)_3(s)$

C $2Al^{3+}(g) + 3SO_4^{2-}(g) \rightarrow Al_2(SO_4)_3(s)$

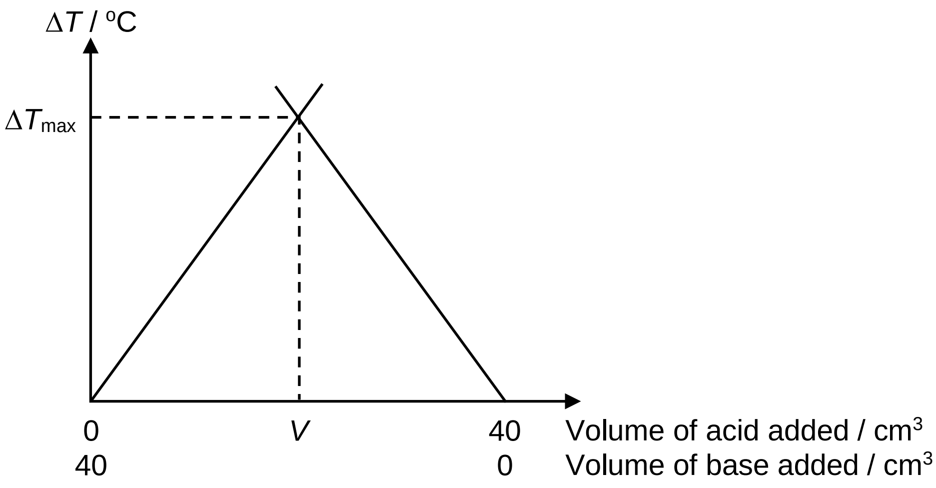
D $2Al^{3+}(g) + 3SO_4^{2-}(g) \rightarrow Al_2(SO_4)_3(g)$

Answer: **A**

Standard enthalpy change of formation of $Al_2(SO_4)_3$ is the enthalpy change when 1 mole of $Al_2(SO_4)_3$ is formed from its elements under standard condition, 1 bar at a specified temperature usually under 298K.

14 Different volumes of HC*l*(aq) and NaOH(aq) were mixed and the change in temperature, ΔT, was measured and the results are plotted in the graph below. ΔT_{max} is the maximum temperature change measured.

Total volume of each reaction mixture was kept constant.
Both acid and base used are of the same concentration.



Which statement is not true about the experiment?

A Before the ΔT_{max}, the acid is the limiting reagent.

B V represents the point when stoichiometric amounts of acid and base react completely with each other.

C When the same concentration of $NH_3(aq)$ is used instead, ΔT_{max} will decrease while V remains the same.

D When the same concentration of $H_2SO_4(aq)$ is used instead, V remains the same while ΔT_{max} increases.

Answer: **D**

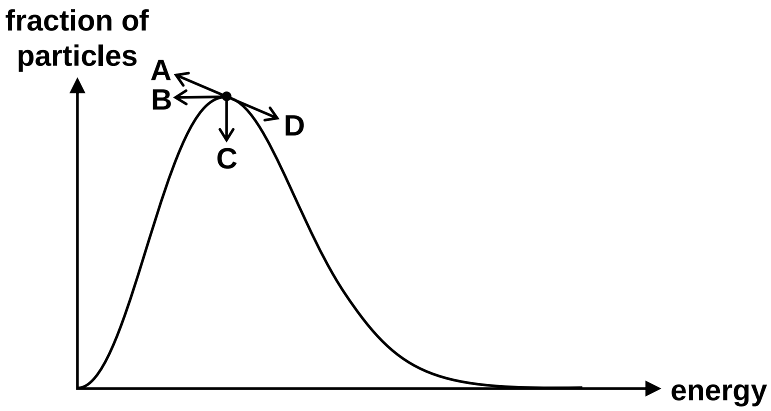
A: True. Before ΔT_{max}, the base is the limiting reagent. Hence the amount of water depends on the amount of base and the ΔT increases as q evolved increases.

B: True. Based on the given concentration of acid and base, V is the volume of complete neutralisation.

C: True. NH_3 is mono-acidic, like NaOH, hence, V will remain the same. NH_3 is a weak base. Hence the heat evolved from neutralisation is not released completely. Some energy is absorb to cause NH_3 to dissociate completely.

D: False. V will increase as H_2SO_4 is dibasic and will require a larger volume of NaOH for complete neutralisation. ΔT_{max} will also increase as the amount of water formed increase, leading to larger heat evolved.

15 The Maxwell-Boltzman distribution curve of a reaction at T K is shown below. How will the peak of the distribution curve change when temperature is lowered to (T–10) K?



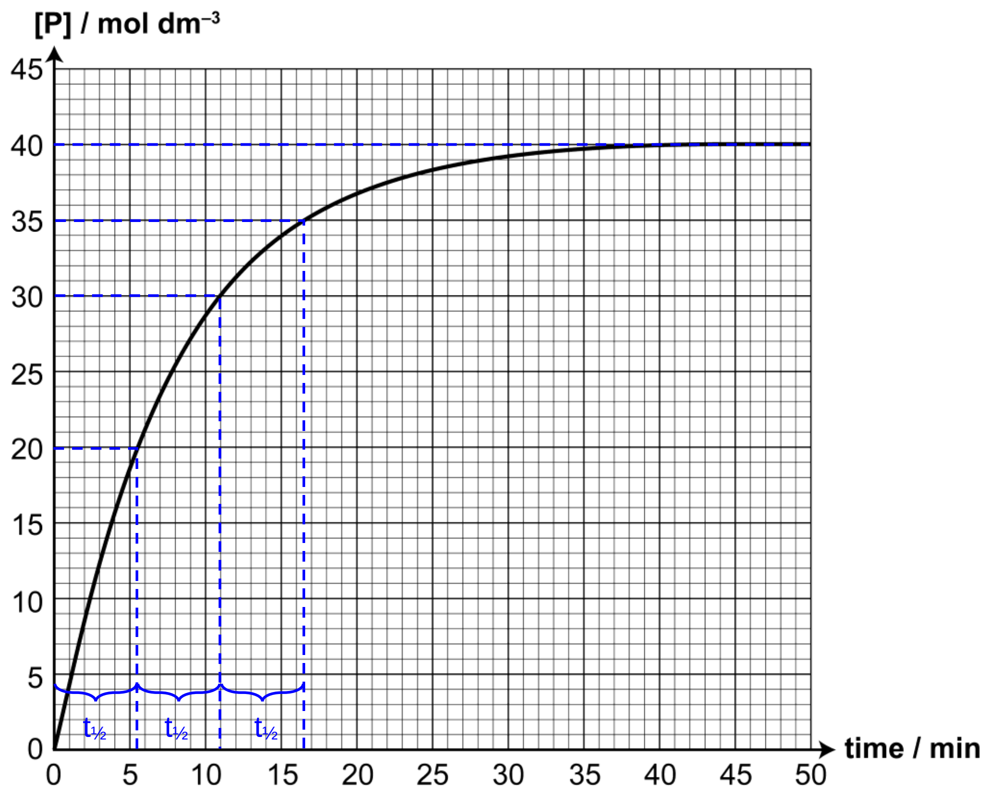
Answer: **A**

When T is decreased, average kinetic energy of particles decreases, thus the distribution curve narrows, sharpens and shifts to the left.

16 Product **P** can be formed by reacting compound **R** under suitable conditions.

$$\mathbf{R} \rightarrow \mathbf{P}$$

In a kinetic study, the formation of product **P** was monitored as shown below.



What conclusions can be drawn from the graph?

1 The half-life of the reaction is about 2.5 min.

2 The rate constant, k, is 0.126 min^{-1}.

3 The reaction is first order with respect to **P**.

A 1 only **B** 2 only **C** 1 and 3 **D** 2 and 3

Answer: **B**

The reaction has a constant half-life of 5.5 min. ⇒ statement 1 is incorrect.

Since $t_{½} = \frac{ln2}{k}$, thus $k = \frac{ln2}{5.5}$ = 0.126 min^{-1} ⇒ statement 2 is correct.

Compound **P** is the product. Order of rxn is w.r.t reactant ⇒ statement 3 is incorrect.

17 Catalyst is often added to speed up the rate of reaction.

Which of the following is not true of a catalyst?

A The activation energy of the reaction is lowered in the presence of a catalyst.

B When a catalyst is added, the rate constant of the reaction is increased.

C A catalyst remains chemically unchanged throughout the reaction.

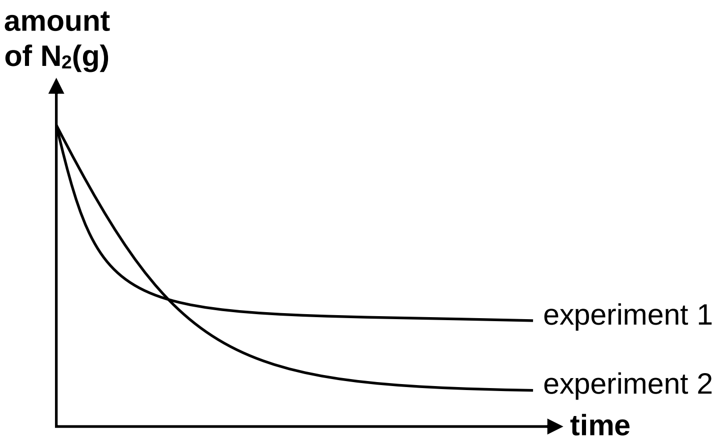
D A catalyst is added in small amounts as it is regenerated at the end of the reaction.

Answer: **C**
Catalyst speeds up the rate of reaction by providing an alternative reaction pathway of lower activation energy. Since activation energy is lowered, a catalyst thus increase rate constant (k = A$e^{-\frac{Ea}{RT}}$). A catalyst participates in the reaction mechanism and may be changed (e.g. homogeneous catalyst Fe^{2+} and Fe^{3+}) in the reaction. It is then regenerated at the end of the reaction , thus only a small amount of catalyst is required.

18 Ammonia gas is produced by reacting nitrogen gas and hydrogen gas and has the equilibrium shown:

$$N_2(g) + H_2(g) \rightleftharpoons NH_3(g) \qquad \Delta H < 0$$

The change in amount of nitrogen was monitored across time for two experiments. Experiment 1 is conducted at 450 ºC and 250 atm while experiment 2 was carried out under a different set of conditions.



What change in conditions would result in experiment 2?

A Finely divided iron catalyst was added.

B Larger amount of nitrogen gas was reacted.

C Pressure of the reaction was lowered to 100 atm.

D Temperature of the reaction was lowered to 350 ºC.

Answer: **D**

From expt 1 to expt 2, the gradient of the graph was less steep, and expt 2 plateaued at a lower point. This shows that position of equilibrium has shifted to the right, (a greater amount of N_2(g) remains at equilibrium) and rate of reaction to achieve equilibrium is slower.

A Incorrect. A catalyst will only speed up the rate in which equilibrium is achieved, it does not shift the position of equilibrium. Addition of catalyst will results in expt 2 having a steeper gradient, equilibrium is reached earlier but plateaus at the same point as expt 1.

B Incorrect. Both graphs started at the same point, thus both experiments started with the same amount of N_2(g) .

C Incorrect as when P is lowered, although rate of reaction decreases, but position of equilibrium shifts left to favour reaction that produces more amount of gases. Thus, expt 2 will have a higher eqm amount of N_2(g) and expt 2 should plateau at a higher point than expt 1.

D Correct. When T decreases, position of equilibrium shifts right to produce more heat given that the forward reaction is exothermic. Thus, expt 2 will have a lower eqm amount of N_2(g). In addition, rate of reaction decreased due to lower temperatures. Thus, expt 2 will have a gentler gradient and plateaus at a lower point than expt 1.

19 Which of the following statements about a reaction at dynamic equilibrium are true?

1 The rate of forward reaction and rate of backward reaction is zero.

2 Equilibrium can only be achieved in a closed system.

3 Equilibrium can be established from either direction.

A 1 only **B** 1 and 2 **C** 2 and 3 **D** 1, 2 and 3

Answer: **C**
The rate of forward reaction and rate of backward reaction is constant and equal, but not zero.

20 Which pair contains an Arrhenius base and a Brønsted acid?

A NH_3 and NH_4^+

B $Ba(OH)_2$ and H_3O^+

C KOH and CH_3COO^-

D Na_2CO_3 and HCl

Answer: B

Arhenius base dissociates OH^- in aqueous medium while Brønsted acid dissociates H^+

A NH_3 and NH_4^+ $\Rightarrow$ NH_3 is not an Arhenius base, it's a Brønsted base

B $Ba(OH)_2$ and $H_3O^+ \Rightarrow Ba(OH)_2$ dissociates to form OH^- and H_3O^+ dissociates to form H^+

C KOH and $CH_3COO^- \Rightarrow CH_3COO^-$ is not a Brønsted acid, it's a Brønsted base

D H_2CO_3 and $HCl \Rightarrow$ Both compounds are Brønsted acid

21 When a strong base is titrated against a weak acid, a drastic increase in pH from pH 8 to 10 is observed

Which indicator would not be suitable for this titration?

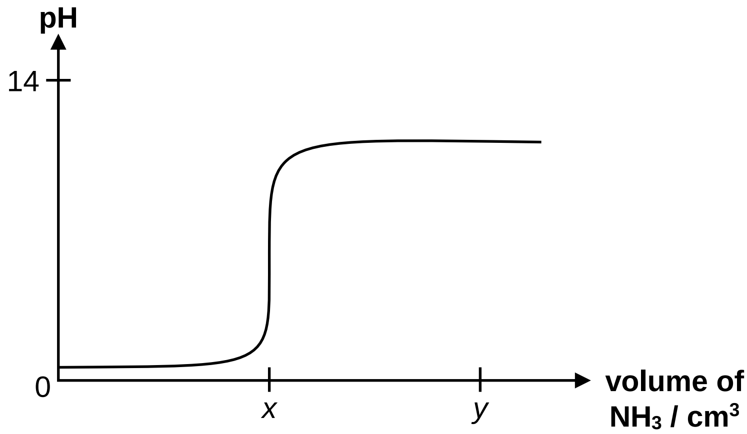
	indicator	Working pH range
A	Litmus	6.0 – 8.0
B	Cresol Purple	7.4 – 9.0
C	Phenolphthalein	8.2 – 10.0
D	Thymolphthalein	9.3 – 10.5

Answer: A

An indicator is suitable for a titration if the working pH range of the indicator (pH range for colour changes) lies within the range of rapid pH change for the titration.

All indicators except litmus indicator has a working pH range that lies within the range of rapid pH change.

22 A 25 cm^3 of 0.1 mol dm^{-3} aqueous sulfuric acid was titrated with aqueous ammonia of the same concentration and the following graph is obtained.
(The graph is not drawn to scale.)



Which of the following statements is true?

A The initial pH of the solution is pH 1.

B Equivalence point for this titration is at pH 7.

C When *y* cm^3 of NH_3 is added, a buffer solution is obtained.

D The volume of NH_3 at *x* is 25 cm^3.

Answer: C

A initial pH = lg[H+] where [H+] = 0.2 mol dm^{-3} since $H_2SO_4 \equiv 2H^+$
= 0.699

B Equivalence point of a strong acid – weak base titration pH < 7.

C When y cm^3 of NH_3 is added, NH_3 is added in excess. Thus, all H_2SO_4 has been reacted to form NH_4^+ salt. In the solution, there is weak base, NH_3, and salt of weak base, NH_4^+, thus an alkaline buffer is formed.

D When x cm^3 of NH_3 is added, complete neutralisation occurs.

$H_2SO_4 \equiv 2NH_3$

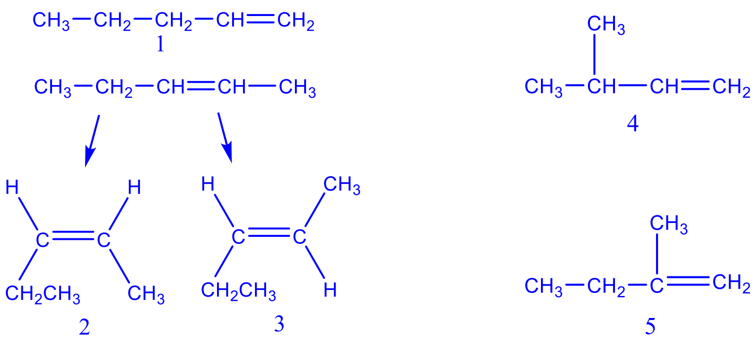
Amt of NH_3 = amt of H^+ = $0.1 \times 2 \times \frac{25}{1000}$ = 0.00500 mol

Vol of NH_3 = 0.00500 mol $\times \frac{1000}{0.1}$ = 50 cm^3

23 What is the total number of constitutional and cis-trans isomers a non-cyclic compound, C_5H_{10}, can have?

A 4 **B** 5 **C** 6 **D** 7

Answer: B



24 A straight chain organic molecule, $CH_2CHCH(OH)CN$, contains a nitrile, –C≡N functional group.

Which of the following regarding the molecule is correct?

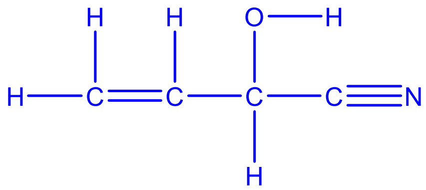
1 It has 10 σ bonds and 3 π bonds.

2 It has an alkene functional group.

3 It can undergo reaction with chlorine gas in the presence of uv light.

A 2 only **B** 1 and 2 **C** 2 and 3 **D** 1, 2 and 3

Answer: D



1: True. There are 10 σ bonds (each covalent bond between atoms has 1 σ bond) and 3 π bonds (1 from C=C, 2 from C≡N).
2: True.
3: True. The H on the C with –OH group (sp3 carbon) can undergo (free radical) substitution.

25 *Use of the Data Booklet is relevant to this question.*

1-bromopropane is reacted separately with the following:

Reagent and condition	**Organic Product**
NaOH(aq), heat	X
NaOH in ethanol, heat	Y
Limited Cl_2 (g), uv light	Z

Rank the products, in terms of relative molecular masses, in ascending order:

	Smallest →		**Largest**
A	X	Y	Z
B	Y	Z	X
C	Y	X	Z
D	Z	Y	X

Answer: **C**
Z: $BrCH_2CH_2CH_2Cl$
X: $CH_2(OH)CH_2CH_3$
Y: $CH_2{=}CHCH_3$

26 In an attempt to produce butanoic acid, a student added excess butan-2-ol to a heated solution of potassium dichromate(VI) and sulfuric acid.

The resultant mixture did not contain any butanoic acid. Why is it so?

A Butan-2-ol is resistant towards oxidation.

B Butan-2-ol forms butan-2-one which cannot be further oxidised.

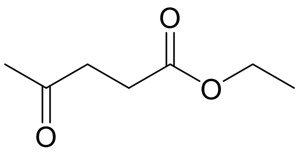
C Potassium dichromate(VI) is not strong enough to fully oxidise alcohol.

D Excess butan-2-ol further react with butanoic acid, resulting in a corresponding ester instead.

Answer: **B**

Butan-2-ol is a secondary alcohol that can be oxidised to form ketone. However, ketone cannot further oxidise to form carboxylic acid.
To form butanoic acid from alcohol, butanal should be used instead.

27 Compound **G**, $C_7H_{12}O_3$, is a fuel additive that reduces soot formation in diesel engines.



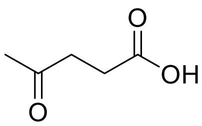
compound **G**

What is correct about compound **G**?

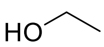
A 19.3 dm^3 of oxygen gas is required for complete combustion of 0.1 mol of **G** at s.t.p.

B 1 mole of **G** reacts 2 moles of hydrogen gas in the presence of nickel catalyst.

C When **G** is heated with acidified potassium manganate(VII), the only organic products formed are:



and



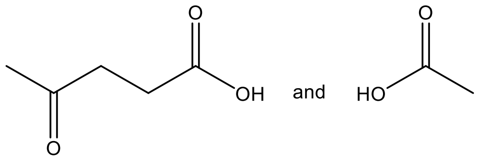
D A constitutional isomer of **G** has only one alcohol functional group and one carboxylic acid group.

Answer: **A**

A: Correct.
$C_7H_{12}O_3 + 8.5O_2 \rightarrow 7\ CO_2 + 6H_2O$
Vol of O_2 = 0.1 x 8.5 x 22.7 = 19.295 =19.3 dm^3

B: wrong. Only the ketone group will be reduced by 1 mole of H_2. Ester will require $LiAlH_4$ for reduction.

C: wrong. The products are:



D: wrong. Esters and carboxylic acid are functional isomers, but ketone are not functional isomers of alcohols

28 Which of the following is a characteristic of thermosetting polymers?

A High tensile strength and high flexibility.

B Heavily branched cross linked polymers.

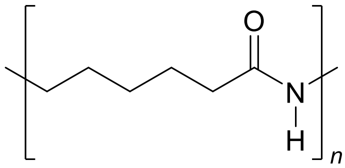
C Linear slightly branched long chain molecules.

D Soften on heating and harden on cooling, can be reused.

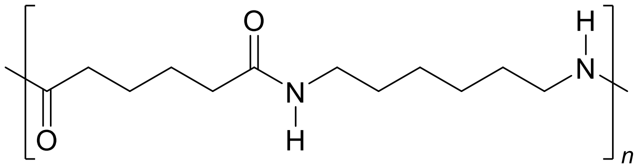
Answer: **B**

Thermosetting polymers with their cross linked structures cannot be remoulded and chars upon heating.

29 Both Nylon 6 and Nylon 66 are polyamides.



nylon 6



nylon 66

Which of the following is true?

A They can be used to make wrinkle free fabric as both polymers are not able to form hydrogen bond between its polymer chains.

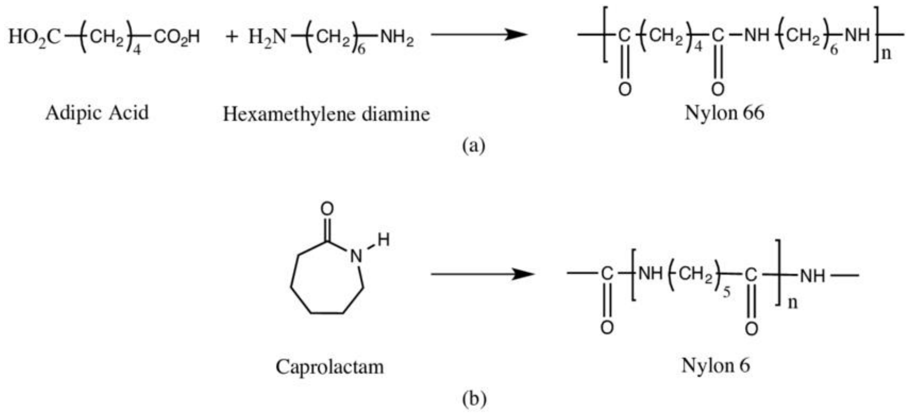
B Both polymers are non-biodegradable as there are non-polar C−C bonds present.

C The formation of nylon 66 results in the elimination of water molecules.

D Both polymers are formed from identical monomers.

Answer: **C**

A: False. Polyamides are able to absorb water by forming hydrogen bonds. Hence when the fabric is not wrinkle free as absorption of water will cause new intermolecular hydrogen bonds to be formed.
B: False. Both are polyamides which have polar C−N bonds which are susceptible to attacks by nucleophiles.
C: True. The condensation between diacid and diamine to form amide bonds results in the elimination of water molecule.
D: False. Both nylon 6 and nylon 66 are made from different monomers:



30 In which contexts is surface area not applicable?

A High tensile strength of graphene

B The ability of a gecko to climb a wall

C Catalytic convertor in automobiles

D Silver nanoparticles in water filtration system

Answer: **A**

ANSWER:

1	B	16	B
2	A	17	C
3	A	18	D
4	C	19	C
5	D	20	B
6	A	21	A
7	C	22	C
8	D	23	B
9	D	24	D
10	B	25	C
11	B	26	B
12	D	27	A
13	A	28	B
14	D	29	C
15	A	30	A